Source-Level Reliability Estimation under Publisher-Disjoint Evaluation

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Abstract. Whether a news outlet can be trusted is conventionally judged by expert organisations, which weigh editorial practice, factual accuracy and transparency at the level of the outlet itself. Those judgements are careful but manual, and the outlets they never reach stay unrated. Computational formulations of the same task usually operate one article at a time: they fit a classifier to pre-labelled items and score it on held-out ones. We instead cast the task as prediction about the source. Our framework has two stages: transformer-based language models rate individual articles, and we then pool those ratings to characterise outlets the models never saw during training. To keep the setting close to deployment, every split is publisher-disjoint, so no outlet contributes to both the training and the test side. We work with 19,476 English-language political news items from 439 outlets carrying NewsGuard ratings. Exact accuracy rises from roughly 0.60 on single articles to 0.69 on outlets, and the vote itself accounts for one to four of those points. Counting a prediction one level away from the true rating as correct, accuracy rises from 0.84 to 0.88. The labels invite that tolerance, because the five levels bin a continuous 0–100 score. We then ask how the errors distribute across the political spectrum, and find systematic associations between the leaning of an outlet and the mistakes made on its articles. We report that disparity as a bound on responsible use. Taken together, the results indicate that text alone carries enough signal to place an unseen outlet on a reliability scale, and we set out what that evidence does and does not license.

Keywords: News source reliability · Publisher reliability inference · Transformer-based text classification · Cross-publisher generalization · Reliability aggregation · Political bias analysis.

1 Introduction

Assessing the reliability of news publishers has become increasingly important for limiting the spread of misleading or low-quality information. Professional organizations such as NewsGuard, the Global Disinformation Index, and Media Bias Fact Check evaluate news outlets according to journalistic standards including transparency, editorial accountability, factual accuracy, and correction practices, and despite methodological differences they exhibit substantial convergence [11,19]. These assessments are performed at the level of the publisher

rather than of individual articles. Trained analysts produce them by hand, so the news outlets they never reach stay unrated.

A large body of computational work instead focuses on article-level classification [1,14,21,24]. However, assessing a publisher across multiple articles can provide a more robust estimate of its reliability than evaluating individual articles in isolation. This approach better reflects the structure of professional assessment, in which reliability is understood as a persistent property of a source: even highly reliable outlets occasionally publish lower-quality content, while unreliable publishers may produce individual articles that appear plausible.

Moreover, article-level evaluation is vulnerable to publisher leakage. When article-level reliability labels are inherited from publisher-level annotations, assessing whether a model can generalize to articles from previously unseen outlets requires publisher-disjoint training and test sets. Some existing studies do not enforce this separation [1,6,23], allowing articles from the same publisher to appear in both sets. Models may therefore exploit publisher-specific patterns instead of learning transferable indicators of reliability [3,26].

We address these limitations by framing news reliability assessment as publisher-level inference. Our two-stage framework uses transformer-based language models to derive a publisher-reliability prediction from each article and then aggregates these predictions across articles to assess previously unseen publishers. We make four main contributions: (i) a strict publisher-disjoint evaluation protocol that prevents source leakage and measures generalization to unseen publishers; (ii) a dataset of 19,476 political news articles from 439 English-language publishers annotated with NewsGuard scores; (iii) evidence that aggregating predictions across articles improves reliability estimation, achieving 0.69 publisher-level accuracy on unseen sources; and (iv) an analysis of how prediction errors are distributed across the political spectrum. Given the potential consequences of such assessments for the outlets being evaluated, Section 6 also articulates the need motivating this work, the conditions under which it may provide benefits, and the harms it may introduce.

2 Related Work

Early work on online news quality focused on article-level classification, detecting misleading content through linguistic features or neural architectures [14], often without considering the publisher.

Publisher reliability and the signals used to estimate it A parallel line shifts the focus to the source, variously termed *reliability*, *trustworthiness* or *credibility* [8, 15,27], whose reputation is known to influence perceived information quality [9]. Computational approaches rely on textual and stylistic features [21] or on external signals such as user interactions, social networks and metadata [4,13,20,28]; journalistic rating systems are commonly used as ground truth [11,19], and recent work explores LLM-based estimation [10,18,24,29].

Datasets and cross-domain generalization Datasets with explicit publisher reliability labels remain limited [1,5,23], and recent work shows that performance degrades under distribution shifts across domains and publishers [3,6,26].

Our contribution In contrast, we focus exclusively on textual content, enforce a strict publisher-disjoint evaluation protocol and report the error disparities the resulting estimates carry.

3 Problem Definition and Methodology

3.1 Problem definition

We formulate the problem as 'News Source Reliability Inference': estimating the reliability level of a publisher from its published content. Because publishers are observed through collections of articles, we decompose it into (i) article-level reliability estimation and (ii) publisher-level inference through aggregation.

Let A be a corpus of news articles, each with a text in $\mathcal{T}$ and a publisher $p(a) \in P$, and let $A_p = \{a \in A \mid p(a) = p\}$ be the articles of p. Each publisher has a fixed reliability level $r_P(p)$ over five ordered categories $L = \{l_1, \ldots, l_5\}$ obtained by binning continuous scores (Section 4); since reliability is a publisher's property, every article inherits it, $r_A(a) = r_P(p(a))$. A political leaning $o_P(p) \in O$, with O ranging over Far Left, Slightly Left, Non labelled, Slightly Right and Far Right, is inherited in the same way. Both inheritance functions are needed because article-level annotation is unavailable; political leaning is never a model input or target, and is retained only for the post-hoc analysis of Section 5.4.

In **Stage 1** we train a classifier $C_{\text{reliability}} : \mathcal{T} \to L$ to predict the reliability of a single article from its text alone. In **Stage 2** we aggregate those predictions through $Agg : L^{|A_p|} \to L$, so that $\hat{r}_P(p) = Agg(\{C_{\text{reliability}}(a) \mid a \in A_p\})$; Section 5.5 instantiates Agg as the mode, i.e. majority voting.

Because each article inherits its publisher's label, articles from the same source are not independently labeled. A random article-level split could therefore place articles from the same publisher in both training and test sets, leaking publisher–label associations across the two partitions. Letting P_{train} and P_{test} denote the sets of publishers represented in the training and test data, respectively, we enforce

$$P_{\text{train}} \cap P_{\text{test}} = \emptyset. \tag{1}$$

All test articles thus originate from publishers that are entirely unseen during training. This protocol reflects the source-level nature of the labels and prevents models from improving test performance by memorizing publisher-label associations. For the same reason, we do not mask source identifiers: because every test publisher is absent from the training data, its identifier is also unseen and cannot serve as a key to a memorized publisher-label association. All test articles thus originate from publishers that are entirely unseen during training. This protocol reflects the source-level nature of the labels and prevents models from improving test performance by memorizing publisher-label associations. For the

same reason, we do not mask source identifiers: because every test publisher is absent from the training data, its identifier is also unseen and cannot serve as a key to a memorized publisher-label association.

3.2 Evaluation Metrics and Baselines

We use the same metrics for both article-level and source-level evaluations.

Standard and Macro Metrics We report Accuracy as the ratio of correct predictions. Because class frequencies differ moderately (l_4 is about 2.6 times larger than l_5), we prefer Macro-averaged Precision, Recall and F1, which treat all five classes equally.

Tolerant Metrics Given the ordinal nature of our labels, standard accuracy may be overly punitive for minor deviations. We therefore adopt *Tolerant Accuracy* [17], which assigns partial credit to predictions falling into adjacent categories. Let N be the total number of samples and c_{ij} the count of instances with true label i predicted as j:

$$\text{Accuracy}_{\text{tolerant}} = \frac{1}{N} \sum_{i \in L} \sum_{j \in L} w_{ij} \cdot c_{ij}, \qquad w_{ij} = \begin{cases} 1 & \text{if } |i-j| = 0 \\ \alpha & \text{if } |i-j| = 1 \\ 0 & \text{if } |i-j| > 1 \end{cases} \tag{2}$$

The parameter $\alpha \in [0, 1]$ controls the tolerance level. We report results for $\alpha = 0.5$ (half credit for adjacent errors) and $\alpha = 1.0$ (full credit, effectively treating adjacent classes as correct).

Baselines We compare against two non-informative baselines and report both of them on every metric. The **Majority Baseline** assigns every instance to the most frequent class observed in the training set, which is the best any constant predictor can do on accuracy; it scores zero Recall and F1 on every other class, so its macro figures are low by construction. The **Uniform Random Baseline** draws each of the five reliability levels with equal probability. Its expected accuracy, Macro Precision and Macro Recall are all 0.20, while its expected Macro F1 is 0.19, slightly lower because the classes are unevenly represented. Being stochastic, it is reported as the mean over 1,000 seeded draws.

3.3 Bias Assessment and Statistical Analysis

Beyond aggregate performance, we investigate *how* and *why* models fail.

Error Analysis We compute the Error Rate per level, $(1 - \text{Recall}_l)$. To identify systematic over- or under-estimation, we analyze the Mean Error Direction (MED) over misclassified samples:

$$\text{MED} = \frac{1}{|E|} \sum_{i \in E} (\hat{y}_i - y_i) \tag{3}$$

where $\hat{y}_i$ and y_i represent the predicted and true reliability levels (mapped to their ordinal values $1, \ldots, 5$), respectively. Positive values indicate a tendency to overestimate reliability, while negative values indicate a tendency to underestimate reliability.

Statistical Significance of Political Bias To determine whether classification errors depend on the publisher's political leaning, we conduct a Pearson's χ^2 test of independence. We test the null hypothesis (H_0) that error frequency is independent of political orientation.

To quantify the magnitude of this effect independent of sample size, we compute Cramér's V:

$$V = \sqrt{\frac{\chi^2}{N \cdot \min(r-1, c-1)}} \tag{4}$$

where N is the total number of samples, while r and c represent the number of rows (political orientations) and columns (classification outcomes) in the contingency table.

4 Dataset and Labeling

Ground Truth: NewsGuard The publisher-level reliability function $r_P : P \to L$ is derived from NewsGuard,[1] a professional media rating system in which trained journalists evaluate online news outlets. Each publisher $p \in P$ is assessed against nine widely adopted journalistic criteria, including the use of deceptive headlines, transparency of ownership and financing, and correction practices.

The evaluation produces a continuous reliability score in $[0, 100]$, measuring a publisher's adherence to professional journalistic standards.[2] The proprietary ratings were licensed for use to the authors of this publication.

Labels and Selecion of Publishers We discretize the continuous score into five ordinal categories $L = \{l_1, \ldots, l_5\}$ following NewsGuard's own thresholds. Table 1 reports the mapping, the initial number of publishers, the actual scraped domains, and the final number of articles per label.

We restrict P to publishers tagged as *Political News and Commentary*, publishing mainly in English and without a paywall. To prevent the corpus from being dominated by highly unreliable publishers, we cap l_1 at 175 publishers, matching l_4; the initial set contains $|P| = 669$ news outlets.

Article Collection Let A denote the set of collected news articles. For each publisher $p \in P$, we scrape articles from its homepage using the `newspaper3k` library[3], collecting at most 200 articles per outlet to ensure a balanced representation across publishers. The scraper is configured to retrieve English-language

[1] https://www.newsguardtech.com/
[2] https://www.newsguardtech.com/ratings/rating-process-criteria/
[3] https://pypi.org/project/newspaper3k/

Label	Score Range	Initial Dom.	Scraped Dom.	Final Art.
l_1	0–39 (Max Caution)	~~1,211~~ → 175	152	4,871
l_2	40–59 (Caution)	127	69	3,016
l_3	60–74 (Credible w/ Exc.)	124	95	4,620
l_4	75–99 (Generally Credible)	175	88	5,041
l_5	100 (High Credibility)	68	35	1,928

Table 1. Reliability labels and NewsGuard score ranges, with the number of domains after filtering, successfully scraped domains, and collected articles.

content only and to prevent duplicates. Scraping did not succeed for all outlets, due to anti-scraping measures and timeouts, so the final publisher count is lower than the original selection (669 vs 439). We retain only articles of at least 100 words, excluding short announcements and advertisements. This procedure yields an initial corpus of 25,822 articles.

Data Curation and Labeling To preserve the original linguistic characteristics of the articles, we apply no text normalization or cleaning. Some domains nonetheless returned localized content, which we filter out with the `langdetect` library[4]. This process removes 6,346 articles and leaves 19,476 English-language articles in the dataset.

Political Leaning Annotation NewsGuard also provides a political leaning annotation for a subset of publishers, inherited at the article level as $o_A(a) = o_P(p(a))$. Reliability levels are unevenly represented across political categories, a structural property of the media ecosystem that must be considered when interpreting model performance and error patterns in the analyses that follow.

5 Experiments

This section presents the experiments and results.

5.1 Experimental Setup

Following the validation protocol described in Section 3 (Equation 1), we use a 5-fold cross-validation scheme with strict separation of publishers between training and test sets. This assesses generalization to articles from previously unseen publishers and approximates a realistic *in-the-wild* deployment. Folds are stratified on publisher reliability, so that each preserves the same reliability distribution.

We select four transformer-based models spanning encoder baselines and decoder-based large language models. **BERT**-base-uncased [7] establishes a lower

[4] https://pypi.org/project/langdetect/

bound comparable with a vast body of prior work [22]. **RoBERTa**-large [12] tests whether optimized training strategies alone improve reliability detection with the architecture unchanged. **Llama 3** 8B [25] represents the shift toward decoder-only generative models, and **Mistral** 24B [16], roughly three times larger, tests whether additional capacity improves detection or whether performance saturates.

Training Strategies and Hyperparameters Encoders (BERT, RoBERTa) were *fully fine-tuned* with AdamW, a learning rate of $2e^{-5}$ and weight decay [7,12]. For the decoders (Llama, Mistral) full fine-tuning was computationally infeasible, so we adopted Low-Rank Adaptation (LoRA) with a learning rate of $2e^{-4}$ in mixed precision.

Each model was trained for 5 epochs per fold, except Mistral, which was trained for 3. This follows a preliminary convergence study (see Supplementary Material [2], which also reports the remaining optimisation and batching details) showing that validation loss minimizes at Epoch 4 for Llama, while Mistral converges within three epochs. We used early stopping on validation loss and a fixed global random seed (42). The maximum sequence length for input processing was set to 256 tokens, except for Mistral, for which it was increased to 1,024 tokens. This tests whether a longer context improves classification.

All fine-tuned models and code will be released to encourage further research. They are withheld during the review process to preserve anonymity, but will be made available in case of acceptance.

5.2 Stage 1: Article-level Reliability Prediction

We first evaluate the article-level predictors of Stage 1.

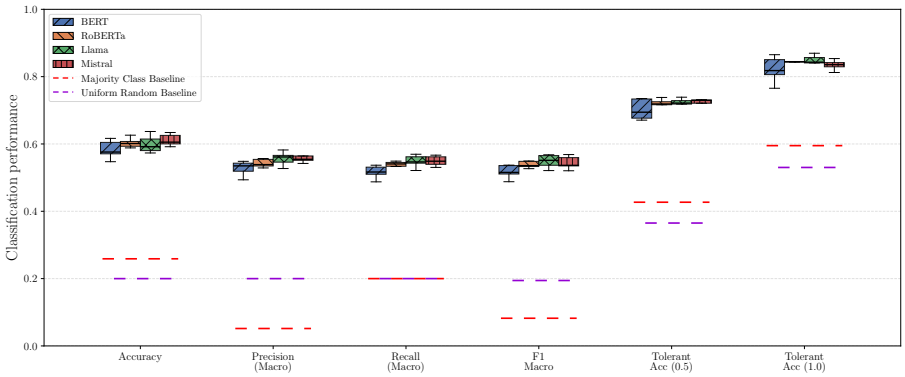


Fig. 1. Article-level metrics across the five folds. Both baselines are drawn across every metric: red for Majority Class, violet for Uniform Random. Where the two coincide, as on Macro Recall at 0.20, the dashes alternate.

Figure 1 illustrates the results, with both baselines drawn across every metric: the Majority Class Baseline (MCB) in red and the Uniform Random Baseline (RB) in violet. All models reach a standard accuracy around 0.60, more than double the MCB, with Mistral highest at 0.61. On macro-averaged performance RoBERTa leads with $F1_{macro}$ of 0.55, followed by Mistral and Llama at 0.54 and BERT at 0.52, all well above the RB of 0.19; the variance across folds is minimal (standard deviation about 0.02). Under the stricter tolerance ($\alpha = 0.5$), Mistral, RoBERTa and Llama reach approximately 0.72 against an adjusted MCB of 0.43. With $\alpha = 1.0$ RoBERTa reaches 0.84 and the others 0.82–0.83; the baseline is naturally higher here (0.60), so the gain is +0.24. Misclassifications are therefore rarely severe, typically involving adjacent labels.

5.3 Biases in Article-level Reliability Prediction

We examine whether classifiers consistently perform better in certain reliability classes than in others.

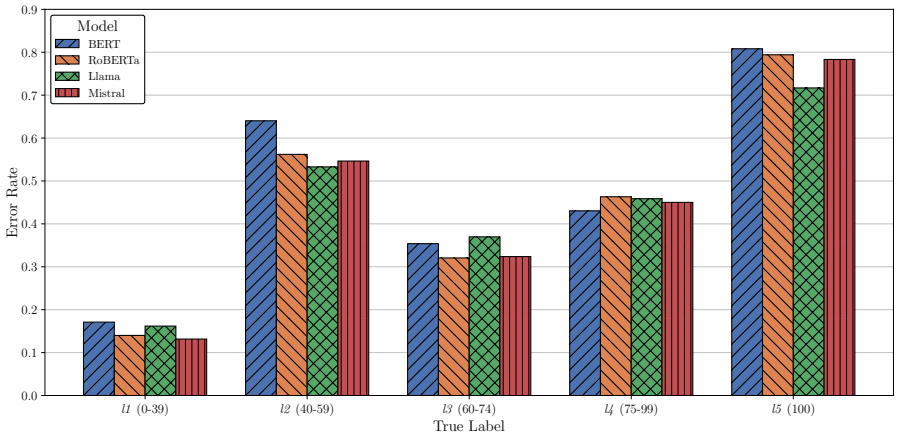


Fig. 2. Per-class error rate, $(1 - \text{Recall}_l)$.

Figure 2 highlights an asymmetry in prediction correctness. The lowest error rates are on l_1 (< 0.20), suggesting that highly unreliable content has recognizable structural features. The highest-reliability articles (l_5) are the most challenging to recognize for the model, with error rates exceeding 70%[5]. Class l_2 also proves difficult, with an error rate of over 50%, while l_3 sits in the middle at approximately 35%.

[5] Class l_5 only includes articles with a reliability score of 100, and is easily confused with l_4, which contains similarly high scores.

Model	Accuracy	χ^2	p-value	Cramér's V
BERT	58.27%	1684.13	$< 10^{-300}$	0.294
RoBERTa	60.34%	1418.31	$< 10^{-300}$	0.270
Llama	59.97%	1207.83	$< 10^{-250}$	0.249
Mistral	61.16%	1343.98	$< 10^{-300}$	0.263

Table 2. Political leaning vs. classification correctness: χ^2 test of independence with p-values, and Cramér's V as effect size.

These patterns remain consistent across all four models, suggesting that the biases are structural properties of the task rather than of a specific architecture; the Supplementary Material [2] reports the error-direction analysis and the full confusion matrices.

5.4 Influence of Political Leaning on Classification Errors

We investigate whether the models treat all sides of the political spectrum equally. Political orientation is a property of the publisher, inherited by each article it publishes (Section 3.1), which enables a post-hoc analysis of how source-level alignment relates to article-level errors.

Error rates are computed separately for each leaning, which makes them independent of class frequency.

All models exhibit systematic variation across political leanings: there are substantially higher error rates for far left and far right outlets (59–72%) compared to lower rates for slightly right and non-labelled outlets (30–34%), and intermediate error rates for slightly left articles (47–58%).

A chi-square test of independence rejects, for every model, the hypothesis that correctness is independent of political leaning (Table 2), with Cramér's V indicating a consistent small-to-moderate effect size. Leaning is a publisher-level attribute, and every article of an outlet carries the same value. We therefore repeat the test with one observation per publisher, and the association holds: χ^2 ranges from 18.8 to 44.8, $p < 10^{-3}$, and V from 0.21 to 0.32. Leaning and reliability are themselves entangled, and no analysis separates them completely: 142 of the 207 slightly right publishers sit in l_1, and no far-right publisher reaches l_5. A regression that controls for the true reliability level still leaves the two extremes with 4.0 to 6.5 times the odds of error of non-labelled outlets, but it rests on thin cells (Supplementary Material [2]). The disparity is therefore a property of the corpus and the model together. A system trained on this ground truth inherits the political structure of the dataset, and the disparity would appear in deployment whatever its mechanism.

We then focus on misclassified instances to assess whether models systematically overestimate or underestimate reliability as a function of political leaning. Figure 3 reports the Mean Error Direction (MED) for each orientation (see Section 3.3); positive values indicate overestimation of reliability, negative values underestimation.

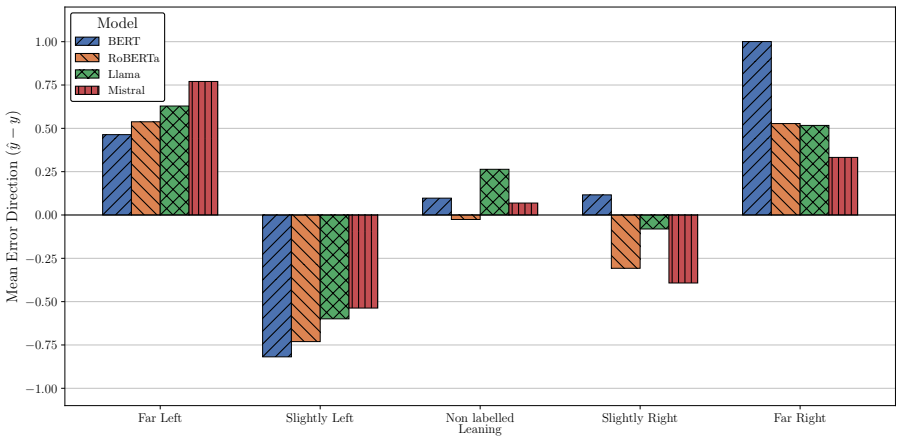


Fig. 3. Mean value and direction of misclassifications, stratified by political leaning. Positive values indicate overestimation (prediction > true label), while negative values indicate underestimation.

Clear directional patterns emerge across all four models. Far Left and Far Right articles exhibit consistent overestimation, with mean signed errors ranging from +0.33 to +1.00, approaching a full reliability level. Slightly Left articles show systematic underestimation, with average values between -0.54 and −0.82. Slightly Right and non-labelled outlets display smaller, less consistent deviations; the Supplementary Material [2] reports the corpus composition by leaning, the per-leaning error rates and the full confusion matrices stratified by political leaning.

5.5 Stage 2: News Source Reliability Inference

Publisher-level inference is the primary objective of this work; the article-level predictors are intermediate. We instantiate *Agg* as the mode of a publisher's article-level predictions, i.e. majority voting, assigning the most frequent class as $\hat{r}_P(p)$, on the hypothesis that combining predictions offsets the noise of individual classifications. Predictions are pooled across the five test folds, so each publisher's reliability is inferred only from articles the model never saw in training.

Figure 4 presents the results against the same two baselines. The majority-class reference differs across the three accuracy variants, since a constant predictor also collects partial and then full credit under tolerance. Under full tolerance the random baseline (0.52) in fact overtakes it (0.50), since a constant predictor earns credit only next to its own level.

With an Accuracy of 0.69 RoBERTa leads, followed closely by Mistral at 0.68, more than triple the Random Baseline. Considering $\text{Accuracy}_{\text{tolerant}}$ (1.0), Llama achieves the top score of 0.88, followed by RoBERTa (0.87) and Mistral (0.86), substantially outperforming the majority class baseline.

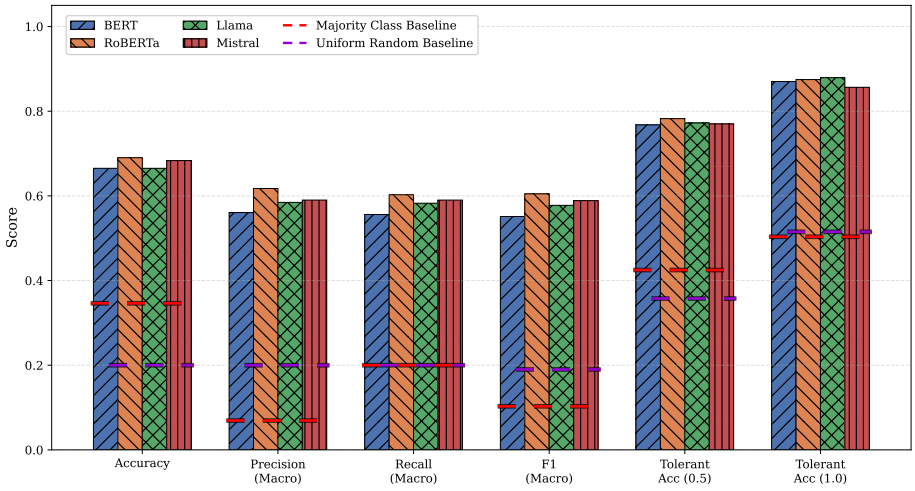


Fig. 4. Publisher-level metrics, with both baselines across every metric. Red: Majority Class Baseline; violet: Uniform Random Baseline. Where they coincide the dashes alternate.

Publisher-level errors are strictly local, concentrating between l_4 (Generally Credible) and l_5 (High Credibility); the Supplementary Material [2] reports full publisher-level confusion matrices and a detailed error analysis.

6 Theory of Change

Work on misinformation is not socially beneficial by default. We therefore state what we expect this work to change, what must hold for it to do so, and how it may cause harm.

The Need and What We Claim About It Expert ratings are mostly manual, and the publishers they never reach stay unrated. That gap motivates automation, but it is not the gap this paper closes. What we claim to change is how the task is posed and evaluated. Reliability is a property of the publisher, yet most computational work predicts it one article at a time, and trains and tests on articles from the same outlets. A model can then recognise the outlet rather than the signal, and what it reports need not hold on a source it has never seen. We pose the task at the publisher level, enforce publisher-disjoint splits, and report what the text alone supports: 0.69 accuracy on unseen outlets (Section 5.5), with errors predominantly local and no gain from model scale.

Preconditions and Their Plausibility Four conditions must hold, and they are unevenly likely to. **The ground truth must be valid**: we inherit the NewsGuard scale together with its editorial judgements, and nothing internal to our

data can test this. **The unit must be the publisher**, never the article: article-level accuracy is 0.60 and supports no claim about an individual piece. **Enough text must be available**. Small and recent outlets lack a rating most often, and they are the least likely to qualify. **The signal must survive beyond our corpus**: our evidence covers English-language political news alone.

Negative Externalities The principal risk is measured rather than hypothesised. Misclassification rates vary systematically with political leaning (Section 5.4): they reach 59–72% for far-left and far-right outlets, against 30–34% for slightly right and non-labelled ones. The models also overestimate reliability at either extreme and underestimate it for slightly left-leaning outlets. Leaning and reliability are entangled in the ground truth itself, so any system trained on it would reproduce the pattern. Two further risks follow from presentation and use. A single level out of five reads as a measurement, although the l_4/l_5 boundary is close to unresolvable (Section 5.3). An inexpensive score is also a target, and surface features are easy to optimise.

What the Evidence Does Not License We measure classification accuracy on held-out publishers, and nothing else. We do not evaluate the approach inside any workflow, we do not measure how it would order outlets for human review, and we do not test abstention on outlets with little text. Any such use needs evidence this paper does not provide. What it does establish is that the error disparity survives clustering at the publisher level and controlling for the true reliability level. Reporting it is part of the result, not an appendix to it.

7 Discussion and Conclusion

In this study, we investigated whether the reliability of news publishers could be inferred from their articles, framing the task as a five-level ordinal classification problem aligned with NewsGuard ratings and evaluating four transformer-based models under a strict publisher-disjoint protocol.

Our findings demonstrate that language models can capture meaningful and generalizable signals of publisher reliability. Interestingly, the substantially larger Mistral 24B does not outperform smaller architectures, suggesting that performance is limited by dataset characteristics rather than model size. Despite operating with a 1,024-token context window, four times larger than the limit used for the other models, Mistral shows no measurable gain, which may indicate that the cues discriminating reliable publishers are concentrated in the beginning of articles and further supports the hypothesis that increasing model size or input length alone cannot overcome limitations imposed by weak supervision. A key finding is that publisher reliability should not be modelled as a property of isolated articles: it emerges as a collective signal that becomes more stable when evidence is aggregated across a source, which supports a shift from article-centric formulations toward source-level inference.

Future research should extend this framework to multilingual settings, explore confidence-aware aggregation strategies, and develop training objectives

tailored to ordinal classification. The value of such systems, however, will not be settled by accuracy alone. Whoever builds one has to show where its errors fall across the political spectrum, which is why we report that disparity alongside the scores.

Disclosure of Interests. The authors have no competing interests to declare.

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